Final Project Report: Railway Flood-Risk Digital Twin Ontology

Course: Knowledge Representation and Semantic Interoperability (KRSI)
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1. Introduction

This project develops an ontology for a **Railway Flood-Risk Digital Twin**, specifically modeling the SNCF Ligne_400 corridor. The ontology semantically integrates infrastructure assets (track segments, culverts, embankments) with hydrological simulation data (rainfall, soil water index), hydraulic results (water surface elevation), and vulnerability assessments (fragility curves). The ultimate goal is to generate RAMS-compliant (Reliability, Availability, Maintainability, and Safety) operational alerts—ranging from standard speed limits to emergency ETCS (European Train Control System) train halts—based on the real-time probability of ballast failure.

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2. LOT Methodology Overview

The ontology was developed following the **Linked Open Terms (LOT)** methodology, an iterative framework for ontology engineering. The applied phases were:

1. **Ontology Requirements Specification:** Defining the purpose, scope, and competency questions.
2. **Ontology Implementation:** Searching for reusable resources, building the conceptual model, and encoding it into OWL.
3. **Ontology Publication:** Evaluating the ontology using reasoners and pitfall scanners, followed by generating documentation using Widoco.

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3. Ontology Development Activities

3.a. Ontology Requirements Specification Document (ORSD)

Field	Description
Purpose	Represent the domain of railway flood-risk assessment, linking physical infrastructure with simulation models and operational alerts.
Scope	Railway assets, terrain profiles, rainfall events, Soil Water Index (SWI), Water Surface Elevation (WSE), failure probabilities, and traffic-light alert verdicts.
Implementation Language	OWL
Intended End-Users	Infrastructure engineers, flood-risk analysts, digital twin developers, train dispatchers.

Field	Description
Intended Uses	Semantic integration of BIM/GIS, automated querying of infrastructure state, and RAMS alert generation.

Selected Competency Questions (CQs)

1. What are the specific types of drainage assets (e.g., culverts, ditches)?
2. What is the ballast elevation (`zBallast`) of a given track segment?
3. Which embankment is adjacent to a specific track segment?
4. What is the computed probability of failure for a specific asset at a given timestep?
5. What operational directive (e.g., "Emergency Halt") does a RED alert status trigger?

3.b. Search for Ontological Resources

The following existing ontologies were selected for reuse:

- **SSN/SOSA:** To represent rain gauges (`sosa:Sensor`) and their rainfall intensity readings (`sosa:Observation`).
- **GeoSPARQL (`geo`):** To represent infrastructure assets as spatial objects (`geo:Feature`).
- **Building Topology Ontology (`bot`):** To represent railway bridges as structural elements (`bot:Element`).
- **FOAF:** To represent the railway operator (e.g., SNCF) as an organization.

3.c. Search for Non-Ontological Resources

Non-ontological resources used to ground the terminology included SNCF internal RAMS standards (for the GREEN/YELLOW/RED alert hierarchy), ISO 55000 asset management terminology, and Wikidata concept URIs (used via `rdfs:seeAlso` annotations to ground classes like "Culvert" to Q1396042).

3.d. Search for Ontology Design Patterns

The **W3C N-ary Relation Pattern** was utilized to model the `AlertVerdict`. Since a verdict links a specific asset, a specific water surface elevation, a computed failure probability, and an operational directive at a specific timestep, it could not be modeled as a simple binary relationship. Instead, `AlertVerdict` was created as a central node linking all these participants.

3.e. Conceptual Model

A conceptual model was designed encompassing 27 classes, 22 object properties, and 18 datatype properties. The core hierarchy splits `InfrastructureAsset` into `TrackSegment`, `DrainageAsset` (subdivided into culverts and ditches), `Embankment`, and `Bridge`. The model strictly enforces disjointness (e.g., an asset cannot be both a track segment and a drainage ditch).

These concepts and properties were formalized in two tabular structures: `Classes.xlsx` and `Properties.xlsx`.

3.f. OWL Implementation

The conceptual model was translated into an OWL/XML file (`RailwayFloodTwin.owl`). Key implementation details include:

- Definition of the namespace URI: `https://w3id.org/def/RailwayFloodTwin#`
- Addition of cardinality constraints (e.g., a RailwayCorridor must contain at least 1 asset).
- Instantiation of named individuals for the `AlertStatus` enumeration (GREEN, YELLOW, RED) and `OperationalDirective` (Standby, Speed Restriction, Emergency Halt), enforced via `owl:AllDifferent` disjointness axioms.

3.g. Evaluation

Reasoner Validation

The ontology was loaded into Protégé and checked with the **HermiT** reasoner.

- **Consistency:** The ontology is logically consistent. No unsatisfiable classes or logical contradictions were found.
- **Inference:** The class hierarchy remained stable post-inference, and individuals were correctly classified.

Protégé Reasoner Output

OOPS! Pitfall Scanner

The ontology implementation (`RailwayFloodTwin.owl`) was evaluated using the **OOPS! (OntOlogy Pitfall Scanner!)** web service. The scan results are summarized below:

- **Critical Pitfalls:** 0
- **Important Pitfalls:** 2 detected (P11 and P35)
  - *P11: Missing domain or range in properties* (on `rft:timestamp`). We left it blank intentionally because `timestamp` is shared across `Observation`, `RainfallEvent`, and `AlertVerdict` to allow reuse, which is a standard ontology design practice.
  - *P35: Untyped property* (on `sosa:madeObservation`). We resolved this in the latest update of `RailwayFloodTwin.owl` by explicitly declaring `sosa:madeObservation` as an `owl:ObjectProperty` in the ontology header.
- **Minor Pitfalls:** 4 detected (P04, P08, P13, P32)
  - *P04: Creating unconnected ontology elements* & *P08: Missing annotations* on external classes (`sosa:Sensor`, `geo:Feature` etc.). These are external classes declared locally for linking and do not require full local definitions.
  - *P13: Inverse relationships not explicitly declared* (some properties do not require a backward relationship for our digital twin logic).
  - *P32: Several classes with the same label* (e.g. `rft:Sensor` and `sosa:Sensor` sharing the label "Sensor", which is standard for local subclasses).
- **Resolution:** The important pitfall P35 was resolved by adding explicit type declarations in the OWL file. All other minor pitfalls are standard side-effects of reusing external W3C/OGC vocabularies.

OOPS! Evaluation Results

3.h. Documentation & Publication

The HTML documentation was generated using the **OnToology** framework (which runs **Widoco** and **LODE** under the hood).

- **Documentation Link:** http://ontoology.linkeddata.es/previews/tinluan/KRSI/latest/OnToology/RailwayFloodTwin.owl/documentation/index-en.html
- **Repository Link:** https://github.com/tinluan/KRSI
- **Method:** The repository was integrated with OnToology to generate documentation and diagrams automatically upon pushing updates. A license annotation (`http://purl.org/NET/rdflicense/cc-by4.0` for CC-BY 4.0) is declared in the OWL header to ensure open terms access.

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4. Sample Data & SPARQL Queries

A subset of real data from the Ligne_400 Digital Twin (20 physical assets extracted from `z_config.json`) was instantiated into the ontology (`RailwayFloodTwin_with_individuals.owl`). A simulated flash-flood event (T=15) was injected, linking a `RainfallEvent` to a `SoilWaterIndex`, calculating `WaterSurfaceElevation`, and finally generating a YELLOW `AlertVerdict` for a culvert (`Buse_0`).

Eight SPARQL queries were developed to demonstrate the ontology's analytical power. For example, querying the n-ary alert verdict to find actionable directives:

```
PREFIX rft: <https://w3id.org/def/RailwayFloodTwin#>
SELECT ?assetId ?wseValue ?statusName ?directiveName
WHERE {
  ?verdict a rft:AlertVerdict ;
           rft:verdictForAsset ?asset ;
           rft:verdictHasWSE ?wse ;
           rft:verdictHasStatus ?status ;
           rft:verdictHasDirective ?directive .

  ?asset rft:assetId ?assetId .
  ?wse rft:wseValue ?wseValue .
  ?status rdfs:label ?statusName .
  ?directive rdfs:label ?directiveName .
}
```

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5. Conclusions

The `RailwayFloodTwin` ontology successfully encapsulates the complex, multi-disciplinary logic required to run a railway flood-risk digital twin. By formalizing the engineering chain—from hydrological rainfall filtering (SWI) to hydraulic depth extraction (HEC-RAS WSE) and ultimately structural vulnerability (Fragility Curves)—the ontology provides a machine-readable foundation for automated ETCS signaling. Adhering to the LOT methodology ensured that the ontology is logically consistent, well-documented, and aligned with external W3C standards (SOSA, GeoSPARQL).